

DRISTI ROY

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EDUCATION

University of California, Los Angeles | B.S. CS at Samueli School of Engineering | Los Angeles, CA | *Class of 2030*

Incoming Freshman, GPA N/A. Calculus IV, CS 31/1, Linguistics 1

Evergreen Valley College | San Jose, CA

UW GPA – 4.0 | Concurrent Enrollment Student

Advanced DSA: Computer Science 2 in C++ / Programming 1 in C

West Valley College | Saratoga, CA

UW GPA – 4.0 | Dual Enrollment Student

Linear Algebra, Advanced Python / Public Speaking / Graphic Design a.k.a Intro to App Design // Upcoming: Linear Algebra / Differential Equations

De Anza College | Cupertino, CA

UW GPA – 4.0 | Concurrent Enrollment Student

Calculus III

Branham High School | San Jose, CA | Class of 2026

GPA – 4.4309 14AP 2H

AP CSP - 5, AP CSA - 5, AP Calc AB, BC, AB Subscore - 5, AP Chem - 5, AP Stats - 5, AP Physics 1, 2, C Mech - 5, AP Lang/Lit - 5

Current: AP Biology, AP Spanish, AP Government, AP Literature, Honors Software Development

UC Scout Online High School | Online Campus

UW GPA – 4.0 | Asynchronous Student, AP Physics 1 - 5 // Upcoming: AP Physics C (Electricity and Magnetism)

RESEARCH EXPERIENCE

AlgoverseAI, LLM Student Researcher | May 2025 - Sept 2025

Selected for partial merit scholarship (~1% of applicants) to study LLM interpretability. Proposed novel mechanistic interpretability framework (combination of sparse autoencoders with linear probes) for behavioral alignment against sycophancy in Gemma with NEFT, presenting to NEMI, NeurIPS Cognitive Interp and ReliableML. "A Few Bad Neurons: Isolating and Surgically Correcting Sycophancy"

I-Lab, Bioinformatics Research Student | Aug 2024

Mentored by Dr. Wang and Dr. Younis from Stanford Microbiology Robinson Lab, combining RNA data samples for overlooked risk factor genes for T2D in nerve tissue. Currently writing the paper. Learned how to conduct BulkRNA Seq/SCRNA Seq Analysis, as well as a few common wet lab techniques like recombinant DNA techniques.

UCSB Summer Research Academy, Researcher (Neuromorphic Computing) | Jun - Aug 2024

Studied how neurons are mathematically and electronically modeled in computers. Also trained spiking neural networks with SNNtTorch. 2% acceptance for track. Wrote reflection discussing the future of neuromorphic computing while researching parallel conversions between different spike coding systems. While suggesting this implementation I learned the basics of HDLs like Verilog and Pyrtl. Presented to UCSB SRA Capstone Seminar, may expand in the future.

SKILLS/PROJECTS

Programming Languages: Python (PyTorch, scikit-learn, pandas, matplotlib, seaborn, SNNtTorch, etc.) and JS: 4 years | R (data analysis, bioinformatics, RNA-Seq pipelines): 3 years | Web stack (HTML/CSS/JavaScript, React): 3 years | Java, C, C++, C#: 2.5 years

Specialized Skills: Machine Learning & Deep Learning | Mechanistic Interpretability (sparse autoencoders, linear probes) | Bioinformatics (BulkRNA-Seq, scRNA-Seq analysis) | Neuromorphic Computing (spiking neural networks, HDLs) | Computer Vision (YOLO, OpenCV) | NLP | Robotics (Arduino, Raspberry Pi, VEX PROS) | CAD (Fusion360) | PCB Design | Firmware Programming

Tools & Frameworks: Git/GitHub, Vercel deployment, GitHub API integration, Model Context Protocol, browser extension development, etc.

Some Projects:

MCP OAuth Automation Tool: Browser app automating Model Context Protocol authentication for non-technical users; streamlines Google Suite MCP setup process

OmniCat: Gesture-Controlled RC Robot: Remote hand gesture controllable RC cat + GitHub assistant. Responsible for part design, all CAD work, and GitHub API integration code

KeyScopeAI: Real-Time Contradiction Detector: AI model deployed as website detecting logical contradictions in live speech. Won AI Award at LingHacks V

Handy Tools: Browser extension for Pomodoro study sessions with time capsule features. Deployed to Vercel for public use

Lab Tool Classification System: YOLO + OpenCV computer vision model for automated trash sorting and lab tool classification. Built for GWC Sustainability challenge

DroydPad: Custom Macropad: 12-switch macropad with custom PCB design. First major experience with PCB design and hardware prototyping

DroydBadge: Custom PCB USB chargeable conference badge.

DataSci Forest Fires Analysis: Statistical analysis project on forest fire patterns, extending AP Statistics coursework with advanced data science techniques
Quadruped Robot Dog (EVODYNE): Autonomous robot dog built during EVODYNE Genesis Track. Learned microcontroller-based robotics and CAD design

ADDITIONAL PROGRAMS & ACTIVITIES

PilotCity Cognixion and Seed Summer Internships | Jun - Aug 2025

Developed the Magic Glove, an RP2040-based wearable with multi-sensor input (flex sensors, magnetometers, OLED, laser) for gesture recognition, and prototyped Cognixion's Speakprose iPhone keyboard by integrating sentiment analysis and eye-tracking features.

ATDP Berkeley Data Science Student | Jun - Aug 2023

Enrolled in Data Science I in R. Transferred Data Analysis skills from Python to R. Developed DS project on COVID diagnosis and risk assessment based on patient demographics, attributes, and CT scan. Wrote weekly reflections/essay on statistical analysis methods.

Inspirit AI Scholar | Jan - May 2023

First exposure to using AI and building neural networks with python general-purpose ML toolkits. Mentored under professor Dr. Kuan-Chuen Wu, built and tested different AI models for NLP in Finance sentiment of stock tweets.

President of BHS Computer Science Club | Sep 2022 - Present

Build projects, run meetings, create slides, discuss CS updates, form hackathon groups to attend competitions, and help with school events such as Blu's Hacks and VEX comps. Currently working with a depth sensor and teaching members data science/research skills, seeking to write a data analysis report on how media censorship in different forms relates to state/country prioritization and success of education.

VEX6374 G-Team Co-Captain and Creative Director | Aug 2023 - Present

Started as Robotics Outreach Team Member, delivering grams to high schoolers with robots. Recently joined as founding member of the all girls competition team, 6374-G, as well as the SciOly robot event. I learned CAD in Fusion360 to design VEX bot for 2024 challenge, I've been building + learning how to code the robot with PROS, I can program/wire microcontrollers such as Arduino/Raspberry Pi, and I discuss robot updates at weekly online meetings.

California Scholarship Federation BHS President, CUHSD SAB and CTE Board | 2022 - Present

Lead biweekly meetings to share service opportunities with 20-30 members yearly and track hours. Member of CUHSD CTE Advisory Board and Student Advisory Board for 2 years.

Public Programming Volunteer | Jun 2023 - Present

I volunteer at the Tech Interactive Museum and Hiller Aviation Museum, where I guide kids through STEM kits. I also volunteer as treasurer, tutor, and writer at my local library.

HONORS & AWARDS

- NeurIPS CogInterp and ReliableML Workshop Acceptance + New England Mechanistic Interpretability Workshop Acceptance (12th)
- National Merit Commended Scholar (top 1.3% in the country) and National School Recognition Award (11th)
- Hack Club Shipwrecked (1 of 130/5000+) & Undercity Invites, LingHacks V AI Award, Blu's Hacks 3rd Place (10th-11th)
- Berkeley Math Tournament Distinguished Honor, Stanford Math Tournament Honor (11th)
- California Scholarship Federation BHS President and Sealbearer, AP Scholar with Distinction, NHS
- Piano 8+ years: Carnegie Hall performance (American Protégé International Winner), seasonal recitals, Certificate of Merit Level 9 with State Honors (5 consecutive years), Advanced level, CWCC Piano 1st Place
- Flute 6+ years: 50+ concert performances, Wind Ensemble Most Valuable Member (9th grade), Music Leadership Team Show Runner (10th-11th), Branham Wind Ensemble/Pep Band member
- Kung Fu: 2x Gold Medalist at Golden State Wushu Championship (qualified for National and Pan-American Invitational), 2x Black Belt Testing Champion, IWSD 1st Degree Black Belt, USA Wushu Kung Fu Federation National Member 2024

LEADERSHIP & WORK EXPERIENCE

Hack Club, US Regional Manager (West Coast) | Nov 2025 - Present

Selected as 1 of 3 US Regional Managers (from 500+ applicants) for outstanding community leadership. Currently mentoring 40 Campfire event leads (projected world's largest high school game jam in 200+ cities). Responsible for creating and overseeing up to 100 hackathons across 8 West Coast states within 6 months, mentoring leads, and ensuring quality standards. Paid position.

Hack Club/Branham HS/Cambrian Ed Foundation, Hackathon Director | Feb 2025 - Present

[Daydream SV; Blu's Hacks; GCCO] Planned and executed hackathons for middle/high school students across SV. Raised and distributed \$8000+ in prizes to hundreds of students.

NeuroSpring, Computer Science Intern | Jan 2025 - Present

Developing apps for efficient data annotation and processing of doctor-patient records for training the Neurospring AI Neurologist's NLP model for deploying in EMTs.

Shaolin Kung Fu Zen, Kung Fu Teaching Assistant | Dec 2023 - Oct 2024

Paid part time role as a kung fu trainer as IWSD Black Belt, leading batches of ~20 students (all ages) to learn traditional Shaolin/Wushu Kung Fu forms and go through training/conditioning curriculum. Volunteer for event planning, judging, and set up in in-house tournaments, examinations, performances, and celebrations.